The Local Harmonious Chromatic Problem*

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Abstract

The harmonious chromatic number of graph G, denoted $h(G)$, is the least number of colors which can be used to color $V(G)$ such that adjacent vertices are colored differently and each color-pair occurs on the vertices of an edge at most once. In this paper, we generalize the above problem to be the local harmonious chromatic problem. The local harmonious chromatic problem restricts that the different color-pair requirement is only asked to be satisfied for every edge within distance d for any vertex.

1 Introduction

In [7], Frank, Harary, and Plantholt defined that a *k-coloring* of a graph G is a map $f : V(G) \longrightarrow C$ such that $|C| = k$. If $f(v) = c_1$ and $f(w) = c_2$ for adjacent v and w, we say that the *color-pair* $\{c_1, c_2\}$ is present at the edge vw. A k-coloring is *line-distinguishing* if no color-pair is present at more than one edge, and the *line-distinguishing chromatic number* of G, denoted $\lambda(G)$, is the minimum k such that G has a line-distinguishing k-coloring. Hopcroft and Krishnamoorthy [10] independently studied the above problem and called $\lambda(G)$ the *harmonious coloring number*. They called the problem to determine whether $\lambda(G) \leq k$ the *harmonious coloring problem* and proved that the harmonious coloring problem is NP-complete.

In [12], Miller and Pritikin added one more condition to these colorings, namely that adjacent vertices receive distinct colors and called the problem *the harmonious chromatic problem*. In the appendix of [10], David S. Johnson already gave a very simple proof of the NP-completeness of the harmonious chromatic problem. In this paper, we follow Miller and Pritikin's definition. Specifically, a *harmonious k-coloring* of graph G is a k-coloring with adjacent vertices receiving different colors and all edges receiving different color-pairs. The *harmonious chromatic number* of G, denoted $h(G)$, is the minimum k for which G has a harmonious k-coloring. Determining whether G can be colored harmoniously by k colors is also NP-complete on trees [5] and interval graphs [2].

In this paper, we generalize the harmonious coloring problem to be the local harmonious chromatic problem. The *local harmonious chromatic problem* restricts the different color-pair requirement only needed to be satisfied for every edge within distance d for any vertex. Formally, a k-coloring is a *d-harmonious coloring* if every color-pair is present at most once for every edge within distance d of any vertex $v \in V(G)$, and the *d-harmonious chromatic number* of G, denoted $h_d(G)$, is the minimum k such that G has a d-harmonious coloring. Clearly, the original harmonious chromatic problem is the n-harmonious chromatic problem if $|V(G)| = n$; or, more precisely, the $\lceil x/2 \rceil$-harmonious chromatic problem if x is the diameter of G.

The rest of this paper is organized as follows. In Section 2, we prove that the 1-harmonious chromatic problem for general graphs is NP-complete. In Section 3, we introduce how to find the d-harmonious chromatic number of paths. In Sec-

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tions 4 and 5, we are concerned with the d-harmonious chromatic number of cycles for any positive integer d. Finally, our concluding remarks are given in Section 6.

2 Local harmonious chromatic numbers of paths

In this paper, we consider only simple graphs G with vertex set $V(G)$ and edge set $E(G)$. For definitions of graph theoretic terms see Harary [9]. Since the harmonious coloring problem is NP-complete, determining whether $h_n(G) \leq k$ is also NP-complete. In the following, we prove that the 1-harmonious chromatic problem, i.e., determining whether $h_1(G) \leq k$, is also NP-complete.

Theorem 1. *The* 1*-harmonious chromatic problem for general graphs is NP-complete.*

In the rest of this section, we shall derive the d-harmonious chromatic number for a path P_n, i.e., $h_d(P_n)$. The following two propositions will be used to derive $h_d(P_n)$.

Proposition 2. *A graph G is d-harmonious coloring if and only if no color-pair appears more than once for every subpath P_{2d+1} in G.*

Proposition 3. *Assume that x is the diameter of G. Then $h_d(G) = h(G)$ for every $d \geq \lceil x/2 \rceil$.*

As mentioned in [7], Frank et al. used a trail on a complete graph to determine the harmonious colors of P_n and C_n. For example, the Eulerian circuit 1,2,3,4,5,1,3,5,2,4,1 of K_5 as shown in Figure 1(a) can be used to color the edges of C_{10} (see Figure 1(b)) and P_{11} (see Figure 1(c)). Accordingly, they derived $\lambda(P_n) = \min\left\{2\left\lceil\sqrt{\frac{n-2}{2}}\right\rceil, 2\left\lceil\frac{1+\sqrt{8n-7}}{4}\right\rceil - 1\right\}$ and $\lambda(C_n) = \min\left\{2\left\lceil\sqrt{\frac{n}{2}}\right\rceil, 2\left\lceil\frac{1+\sqrt{8n+1}}{4}\right\rceil - 1\right\}$. In the following, we use $\mathrm{lec}(G)$ to stand for the length of an Eulerian subgraph which has the maximum possible number of edges in $E(G)$. It is obvious that $\mathrm{lec}(K_n) = \binom{n}{2}$ if n is odd; and $\binom{n}{2} - \frac{n}{2}$ otherwise.
By using a similar technique, Miller and Pritikin [12] proposed the following theorem for calculating $h(P_n)$.

Theorem 4 (Miller and Pritikin [12]). $h(P_n) =$
$$\begin{cases} 2k \text{ for all } \mathrm{lec}(K_{2k-1}) < n-1 \leq \mathrm{lec}(K_{2k}) + 1 \\ 2k+1 \text{ for all } \mathrm{lec}(K_{2k}) + 1 < n-1 \leq \mathrm{lec}(K_{2k+1}). \end{cases}$$

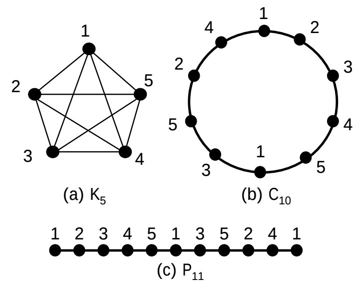

Figure 1: Coloring paths or cycles with an aided Euler circuit.

We can also use a similar technique to find $h_d(P_n)$ as the following theorem.

Theorem 5. *If $n \leq 2d+1$, then $h_d(P_n) = h(P_n)$; otherwise,* $h_d(P_n) =$
$$\begin{cases} 2k \text{ for all } \mathrm{lec}(K_{2k-1}) < 2d \leq \mathrm{lec}(K_{2k}) \\ 2k+1 \text{ for all } \mathrm{lec}(K_{2k}) < 2d \leq \mathrm{lec}(K_{2k+1}). \end{cases}$$

3 Periodical tables

In this section, we are concerned with the d-harmonious chromatic number of C_n. First we define some terms which will be used later. A graph is *pancyclic* if it has cycles of every length ranging from 3 to n [1, 3, 8, 13]. Clearly, K_n is *pancyclic*. We extend the definition of pancyclic to x-pancircuitous. A graph is an *x-pancircuitous graph* if it has circuits of every length ranging from 3 to x. The reason why adding an x before pancircuitous is that, even for K_n, there are no $\mathscr{T}_{\mathrm{lec}(K_n)-1}$ and $\mathscr{T}_{\mathrm{lec}(K_n)-2}$ in K_n if n is an odd number. If n is even, then there do not exist circuits of every length ranging from $\binom{n}{2} - \frac{n}{2} + 1$ to $\binom{n}{2}$. To emphasize that $C_{\mathrm{lec}(K_n)-1}$ and $C_{\mathrm{lec}(K_n)-2}$ cannot be d-harmoniously colored by n colors, these two numbers $\mathrm{lec}(K_n) - 1$ and $\mathrm{lec}(K_n) - 2$ are called *the first n-pump number* and *the second n-pump number*, respectively, for odd integer $n \geq 5$ (or *n-pump numbers* for short). For example, numbers 8 and 9 are the second and the first, respectively, 5-pump numbers while 19 and 20 are the second and the first, respectively, 7-pump numbers. We use the following proposition to describe the absent circuits in K_n when n is odd.

Proposition 6. *If n (≥ 5) is an odd integer, then K_n has no $\mathscr{T}_x$ where x is the first or the second n-pump number.*

Note that if a graph is x-pancircuitous, $x > 3$, then it must be $(x-1)$-pancircuitous. However, by Proposition 6, it might not be $(x+1)$-pancircuitous even if $x + 1 < \binom{n}{2}$. The *pancircuit number* of G, denoted $\mathrm{pn}(G)$, is the maximum k such that G is k-pancircuitous. For integer $d \geq 1$, a clique K_k is said to be the *critical clique* with respect to d, abbreviated as *d-critical clique*, if it is the clique with the smallest k such that K_k has $\mathscr{T}_{2d+1}$. In the rest of this paper, we use K_α to denote the d-critical clique. Lemma 7 describes the value for $\mathrm{pn}(K_n)$ while Lemma 8 gives a necessary and sufficient condition for a d-critical clique which is not $(2d+1)$-pancircuitous.

Lemma 7. *For a complete graph K_n, $n \geq 4$,*

$$\mathrm{pn}(K_n) = \begin{cases} \mathrm{lec}(K_n) - 3 & \text{if } n \text{ is odd} \\ \mathrm{lec}(K_n) & \text{if } n \text{ is even.} \end{cases}$$

Lemma 8. *For a d-critical clique K_α with $d \geq 2$, K_α is not $(2d+1)$-pancircuitous if and only if $\alpha = (1 + \sqrt{16d + 9})/2$ is an odd number.*

Since there exist pump numbers, $h(C_n)$ cannot be computed exactly the same as the computation of $h(P_n)$. After taking into consideration of pump numbers, we can obtain the following theorem.

Theorem 9. *If $\mathrm{lec}(K_{2k-1}) < n \leq \mathrm{lec}(K_{2k}) + 1$, then $h(C_n) = 2k$; otherwise, $h(C_n)$ is equal to*
$$\begin{cases} 2k+1 & \text{if } n \text{ is not a } (2k+1)\text{-pump number,} \\ 2k+2 & \text{if } n \text{ is a } (2k+1)\text{-pump number.} \end{cases}$$

Lemma 10. *For $3 \leq n \leq 4d-1$, $h_d(C_n) = h(C_n)$.*

Corollary 11. *For $d \geq 2$ and $2d \leq n < 4d - 1$, $|h_d(C_{n+1}) - h_d(C_n)| \leq 1$.*

Lemma 12. *If K_α is the d-critical clique, then $\mathrm{lec}(K_\alpha) \leq 4d - 1$.*

By Lemma 10, finding $h_d(C_n)$, for $n \geq 4d$, can be transformed to the problem of determining whether n can be composed of numbers x with $2d \leq x \leq 4d - 1$ such that the maximum $h_d(C_x)$ among all composition numbers is minimum. For brevity, when saying the length of a cycle is composed of the lengths of another two cycles, we simply say that a cycle can be composed of another two cycles. For ease of description, we define a periodical table as follows. Given an integer d, a *periodical table* with respect to d, denoted PT_d, is a table of $2d$ rows and an infinite number of columns. Entry (i, j) of PT_d, denoted $\mathrm{PT}_d(i, j)$, contains cycle C_n with $n = 2jd + i - 1$. Since every $n = 2jd + i - 1$ with integers $j > 1$ and $1 \leq i \leq 2d$ can be expressed as $n = (j-1)2d+(2d+i-1)$, n can be composed of $2d$ and $2d+i-1$. This gives a trivial upper bound of $h_d(C_n) \leq \max\{h_d(C_{2d}), h_d(C_{2d+i-1})\}$ for $n \geq 4d$. To find the exact value of $h_d(C_n)$ for $n \geq 4d$, we need to find a composition of n which results in the exact value of $h_d(C_n)$.

For example, see Table 1 for illustrating a periodical table with $d = 3$. The entries in the first column of Table 1 are $\mathrm{PT}_3(1, 1)$, $\mathrm{PT}_3(2, 1)$, $\ldots$, $\mathrm{PT}_3(6, 1)$ which contain $C_6, C_7, \ldots, C_{11}$, respectively. By Lemma 10, all $h_3(C_n) = h(C_n)$ for $6 \leq n \leq 11$ which are shown in the first column of Table 1. Since all of numbers $n = 12, 18, \ldots$, are multiples of $2d(= 6)$, $h_3(C_n) = 5$ for $n = 12, 18, \ldots$. Number 14 can be composed of by using only 7, i.e., $14 = 7 + 7$. Then, by repeating the colors for C_7 twice on C_{14}, we can obtain $h_3(C_{14}) = h_3(C_7) = 5$. Note that 14 is also equal to $8 + 6$; however, in this composition, $\max\{h_3(C_8), h_3(C_6)\} = \max\{6, 4\} = 6$ which is not the smallest value for $h_3(C_{14})$. The other compositions for cycles in column 2 of Table 1 are $13 = 7+6$, $15 = 9+6$, $16 = 10+6$, and $17 = 10+7$ whose corresponding $h_3(C_n)$ are shown in the second column of Table 1. Note that 15 can be composed either of $9 + 6$ or $8 + 7$; however, both of these two compositions result in $h_3(C_{15}) = 6$. In the third column of Table 1, $21 = 7 + 7 + 7$ and $h_3(C_{21}) = 5$. The other values of n in column 3, i.e., $\mathrm{PT}_3(i, 3)$ with $n = 2 \cdot 3 \cdot d + i - 1$, can be expressed as $n = 2d+(2{\cdot}2{\cdot}d+i-1)$, which is a composition of $2d(= 6)$ and $(2 \cdot 2 \cdot d + i - 1)$. Note that $2 \cdot 2 \cdot d + i - 1$ represents the entry $\mathrm{PT}_3(i, 2)$ which is the previous entry in the same row of $\mathrm{PT}_3(i, 3)$. Similarly, we can obtain $h_3(C_n)$ for cycles in column 4 of Table 1. Note that, by adding $2d$ to the index of each cycle in the fourth column, we can obtain the cycles in the fifth column of Table 1. Thus, their $h_d(C_n)$ can also be found. Similarly, we can find $h_d(C_n)$ for the remaining cycles. From the above analysis, we can find $h_3(C_n)$ for $n \geq 3$ as the following theorem.

Theorem 13.

$$\textit{For } n \geq 3, h_3(C_n) = \begin{cases} n & \text{if } n = 3, 4 \\ 6 & \text{if } n = 8, 9, 11, 15 \\ 5 & \textit{otherwise.} \end{cases}$$

Note that $\mathrm{PT}_1(1, 1)$ results in $n = 2$ and no cycle of length 2. This reveals that the case where $d = 1$ is a special case in finding $h_d(C_n)$ by using the periodical table. Therefore, we consider this case separately in the following theorem and assume that $d \geq 2$ in the rest of this paper.

Table 1: Periodical table $h_3(C_n)$ for $6 \le n \le 29$

$h_3(C_6) = 5$	$h_3(C_{12}) = 5$	$h_3(C_{18}) = 5$	$h_3(C_{24}) = 5$
$h_3(C_7) = 5$	$h_3(C_{13}) = 5$	$h_3(C_{19}) = 5$	$h_3(C_{25}) = 5$
$h_3(C_8) = 6$	$h_3(C_{14}) = 5$	$h_3(C_{20}) = 5$	$h_3(C_{26}) = 5$
$h_3(C_9) = 6$	$h_3(C_{15}) = 6$	$h_3(C_{21}) = 5$	$h_3(C_{27}) = 5$
$h_3(C_{10}) = 5$	$h_3(C_{16}) = 5$	$h_3(C_{22}) = 5$	$h_3(C_{28}) = 5$
$h_3(C_{11}) = 6$	$h_3(C_{17}) = 5$	$h_3(C_{23}) = 5$	$h_3(C_{29}) = 5$

Theorem 14.

$$h_1(C_n) = \begin{cases} 3 & \text{if } n = 3i, \text{ for any integer } i \ge 1 \\ 5 & \text{if } n = 5 \\ 4 & \text{otherwise.} \end{cases}$$

Lemma 15 describes a basic property on the composition of cycle lengths in periodical tables.

Lemma 15. *Let C_x be the cycle in $\mathrm{PT}_d(i,j)$ for integers $i, j \ge 1$. Then, there exist a pair of cycles C_y and C_z which are in $\mathrm{PT}_d(m,1)$ and $\mathrm{PT}_d(i-m+1, j-1)$, respectively, for any $1 \le m \le i$, such that $x = y + z$.*

4 Local harmonious chromatic numbers of cycles

Based on the periodical table, we shall discuss the composition of number n for $n \ge 4d$ under the d-harmonious constraint in order to find $h_d(C_n)$. Recall that K_α is the d-critical clique. By Corollary 11, the possible values of $h_d(C_{2d})$ is $\alpha - 1$, α, and $\alpha + 1$. According to the possible values of $h_d(C_{2d})$ and the parity of α, we classify periodical tables with respect to d into the following eight classes:

(1) $h_d(C_{2d}) = \alpha - 1$ and $2d + 1$ is the second $(\alpha - 1)$-pump number,

(2) $h_d(C_{2d}) = \alpha - 1$, α is even, and $\mathrm{lec}(K_{\alpha-1}) = 2d$,

(3) $h_d(C_{2d}) = \alpha - 1$ and α is odd,

(4) $h_d(C_{2d}) = \alpha$ and $2d + 1$ is the first $(\alpha - 1)$-pump number,

(5) $h_d(C_{2d}) = \alpha$, α is even, and $2d + 1$ is not the first $(\alpha - 1)$-pump number,

(6) $h_d(C_{2d}) = \alpha$, α is odd, and $2d + 2$ is not the second α-pump number,

(7) $h_d(C_{2d}) = \alpha$ and $2d + 2$ is the second α-pump number,

(8) $h_d(C_{2d}) = \alpha + 1$ and α is odd,

which are denoted by $\mathrm{PT}_d^1, \mathrm{PT}_d^2, \ldots, \mathrm{PT}_d^8$, respectively. Note that the case where $h_d(C_{2d}) = \alpha + 1$ with even α is impossible since $h_d(C_{2d}) = \alpha + 1$ must be a pump number which exists only for odd α.

We use the following propositions to describe some properties of these eight classes of periodical tables in which we assume that K_α is the d-critical clique.

Proposition 16. *For the cycles in the first column of a $\mathrm{PT}_d \in \mathrm{PT}_d^1$, only $h_d(C_{2d}) = h_d(C_{2d+3}) = \alpha - 1$. All of the other cycles, say C_n, in the first column of PT_d have $h_d(C_n) \ge \alpha$.*

Proposition 17. *For the cycles in the first column of a $\mathrm{PT}_d \in \mathrm{PT}_d^2 \cup \mathrm{PT}_d^3$, only $h_d(C_{2d}) = \alpha - 1$. All of the other cycles, say C_n, in the first column of PT_d have $h_d(C_n) \ge \alpha$.*

Proposition 18. *For the cycles in the first column of a $\mathrm{PT}_d \in \mathrm{PT}_d^4$, only $h_d(C_{2d+2}) = \alpha - 1$. All of the other cycles, say C_n, in the first column of PT_d have $h_d(C_n) \ge \alpha$.*

Proposition 19. *For the cycles in the first column of a $\mathrm{PT}_d \in \mathrm{PT}_d^5 \cup \mathrm{PT}_d^6 \cup \mathrm{PT}_d^7$, all cycles, say C_n, in the first column of PT_d have $h_d(C_n) \ge \alpha$. Furthermore, there are cycles whose lengths are α-pump numbers in the first column of a $\mathrm{PT}_d \in \mathrm{PT}_d^6 \cup \mathrm{PT}_d^7$.*

Proposition 20. *For the cycles in the first column of a $\mathrm{PT}_d \in \mathrm{PT}_d^8$, only $h_d(C_{2d+1}) = \alpha$. All of the other cycles, say C_n, in the first column of PT_d have $h_d(C_n) \ge \alpha + 1$.*

In a periodical table, we define that the *ith-level leap path* for $i \ge 1$, is a Manhattan path which starts from the edge separating $C_{2\cdot\mathrm{lec}(K_{\alpha+i-1})}$ and $C_{2\cdot\mathrm{lec}(K_{\alpha+i-1})+1}$, then passes through the edges separating $C_{j\cdot\mathrm{lec}(K_{\alpha+i-1})}$ and $C_{j\cdot\mathrm{lec}(K_{\alpha+i-1})+1}$, for $j > 2$, one by one, until it arrives the bottom edge of the table. Note that, once no $C_{j\cdot\mathrm{lec}(K_{\alpha+i-1})}$ exists in the second column of PT_d, the ith-level leap path only contains the bottom edge of the table which is the last leap path. For convenience, we say that the top edge of a periodical table is the *0th-level leap path*. Therefore, there are at least two leap paths, i.e., the top and bottom edges, in a periodical table.

Based on leap paths, we define leap cycles as follows. A cycle C_n with $n \ge 4d$ is called a

(d,i)-*leap cycle*, for $i \geq 0$, if C_n is located between the ith-level and the $(i+1)$th-level leap paths in PT_d. Formally, a cycle C_n in $\mathrm{PT}_d(i,j)$ is a $(d,0)$-leap cycle if and only if $j \geq 2$ and $1 \leq i \leq \min\{2d, j \cdot (\mathrm{lec}(K_\alpha) - 2d) + 1\}$, and is a (d,x)-leap cycle, for $1 \leq x \leq k-2$, if and only if $j \geq 2$ and $j \cdot (\mathrm{lec}(K_{\alpha+x-1}) - 2d) + 1 < i \leq \min\{2d, j \cdot (\mathrm{lec}(K_{\alpha+x}) - 2d) + 1\}$ where k is the number of leap paths. A (d,x)-leap cycle C_n, for $x \geq 0$, in $\mathrm{PT}_d(i,j)$ is called a (d,x)-*pump cycle* if $n = j \cdot \mathrm{lec}(K_{\alpha+x}) - y$, for $y = 1$ or 2, and $\mathrm{lec}(K_{\alpha+x}) - y$ is a $(\alpha+x)$-pump number. A $(d,0)$-leap cycle C_n is called a *type I ripple cycle* if it is in a $\mathrm{PT}_d \in \mathrm{PT}_d^7$ and $n - 2d$ is the first α-pump number. A $(d,0)$-leap cycle C_n is called a *type II ripple cycle* if it is in a $\mathrm{PT}_d \in \mathrm{PT}_d^8$ and C_n is not a $(d,0)$-pump cycle. Since all cycles have to be composed of the cycles in the first column of a PT_d, the cycles C_n in the first column of PT_d with $h_d(C_n) = \alpha + x$, for $x \geq 0$, are called (d,x)-*fundamental cycles*. For simplicity, we also classify cycles C_n in the first column of a PT_d with $h_d(C_n) = \alpha - 1$ into $(d,0)$-*fundamental cycles*. Lemma 21 can be used to determine whether there exist (d,x)-leap cycles or not for a given d and $x \geq 1$.

Lemma 21. *Assume that K_α is the d-critical clique and $x(\geq 1)$ is a positive integer. If $\mathrm{lec}(K_{\alpha+x-1}) < (6d-1)/2$, then there exist (d,x)-leap cycles.*

In the following, Lemmas 22∼29 concern with the d-harmonious chromatic number of $(d,0)$-leap cycles and Lemmas 30 and 31 concern with the d-harmonious chromatic number of (d,x)-leap cycles for PT_d^x, $x = 1, 2, \ldots, 8$.

Lemma 22. *For a $\mathrm{PT}_d \in \mathrm{PT}_d^1$, a $(d,0)$-leap cycle C_n, which is in $\mathrm{PT}_d(i,j)$, has $h_d(C_n) = \alpha - 1$ if and only if it has $i = 3x+1$ for $x = 0, 1, \ldots, \lfloor 2d - 1/3 \rfloor$.*

Lemma 23. *For a $\mathrm{PT}_d \in \mathrm{PT}_d^2 \cup \mathrm{PT}_d^3$, a $(d,0)$-leap cycle C_n in PT_d has $h_d(C_n) = \alpha - 1$ if and only if $n = 2dj$ for $j \geq 2$.*

Lemma 24. *For a $\mathrm{PT}_d \in \mathrm{PT}_d^4$, a $(d,0)$-leap cycle C_n has $h_d(C_n) = \alpha - 1$ if and only if $n = (2d+2)j$ for $j \geq 2$.*

Lemma 25. *Assume that K_α is the d-critical clique and C_n is a $(d,0)$-leap cycle of $\mathrm{PT}_d \in \bigcup_{1 \leq x \leq 5} \mathrm{PT}_d^x$. If $h_d(C_n) \neq \alpha - 1$, then $h_d(C_n) = \alpha$.*

Lemma 26. *For a $(d,0)$-leap cycle C_n of $\mathrm{PT}_d \in \mathrm{PT}_d^6$, if C_n is a $(d,0)$-pump cycle, then $h_d(C_n) = \alpha + 1$; otherwise, $h_d(C_n) = \alpha$.*

Proposition 27. *For a $(d,0)$-leap cycle C_n of $\mathrm{PT}_d \in \mathrm{PT}_d^7$, C_n is a type I ripple cycle if and only if C_n is in $\mathrm{PT}_d(4j-4, j)$ for $2 \leq j \leq \lfloor (2d+5)/4 \rfloor$.*

Lemma 28. *For a $(d,0)$-leap cycle C_n of PT_d in PT_d^7, if C_n is a $(d,0)$-pump cycle or a type I ripple cycle for $j \geq 2$, then $h_d(C_n) = \alpha + 1$; otherwise, $h_d(C_n) = \alpha$.*

Lemma 29. *For a $(d,0)$-leap cycle C_n of PT_d in PT_d^8, if $n = (2d+1)j$, for $j \geq 2$, then $h_d(C_n) = \alpha$; otherwise, $h_d(C_n) = \alpha + 1$.*

By using a similar argument as Lemma 26 for proving the d-harmonious chromatic number of leap cycles, we can obtain Lemmas 30 and 31.

Lemma 30. *Assume that K_α is the d-critical clique. If C_n is a (d,x)-leap cycle, for $x \geq 1$, which is not a (d,x)-pump cycle, then $h_d(C_n) = \alpha + x$.*

Lemma 31. *Assume that K_α is the d-critical clique. If C_n is a (d,x)-pump cycle for $x \geq 1$, then $h_d(C_n) \geq \alpha + x + 1$.*

For convenience to summarize our results, a (d,x)-leap cycle for $x \geq 0$, is called a (d,x)-*perfect cycle* if $h_d(C_n) = \alpha + x - 1$, is called a (d,x)-*critical cycle* if $h_d(C_n) = \alpha + x$, and is called a (d,x)-*pump cycle* if $h_d(C_n) = \alpha + x + 1$. Note that perfect cycles can only exist in $(d,0)$-leap cycles which are described in Lemmas 22∼ 24, critical cycles are described in Lemmas 25∼ 30, and the existence of (d,x)-pump cycles is described in Lemma 21. We summarize our results as the following theorem.

Theorem 32. *Assume that K_α is a d-critical clique with $d \geq 2$. For any C_n, if C_n is a (d,x)-pump cycle or types I and II ripple cycles, then $h_d(C_n) = \alpha + x + 1$; otherwise, $h_d(C_n)$ is equal to*

$$\begin{cases} h(C_n) \text{ if } 3 \leq n \leq 4d-1 \\ \alpha - 1 \text{ if } C_n \text{ is a } (d,0)\text{-perfect cycle} \\ \alpha + x \text{ if } C_n \text{ is a } (d,x)\text{-critical cycle.} \end{cases}$$

5 Conclusion

Here we want to point out that $h_d(C_n)$ might be much larger than $\alpha + 2$ where K_α is a d-critical clique. For example, K_{65} is a 1000-critical clique, namely $d = 1000$ and $\alpha = 65$. Since $4d - 1 = 3999$, $h_{1000}(C_{3999}) = h(C_{3999}) = 91 \geq \alpha + 2$. Even though finding $h_1(G)$ for general graphs is NP-complete, however, for a lot of classes of graphs, there exist polynomial time algorithms for finding their $h_1(G)$. For example, $h_1(T) = \Delta + 1$ where T is a tree. Accordingly, it is interesting to find the value of $h_2(G)$ for trees and other classes of graphs.

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